# Practice workbook · Binary to denary and back (unsigned, 0 to 255)

Converting a whole number 0-255 to an 8-bit unsigned pattern, or an unsigned pattern (including a short sample value) to denary.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-01

# 1. Question 2 · part d.i · 2 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part d.i only.

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**2** Computers manipulate binary patterns. People interpret those patterns.

(a) Complete the table by adding the hexadecimal notation for each of the denary values.

**(2)**

| Denary | Hexadecimal |
|---|---|
| 8 | |
| 12 | |

(b) Identify the expression to give the number of unique binary patterns that can be stored in six bits.

**(1)**

- ☒ **A** $6^2$
- ☒ **B** $6 \times 2$
- ☒ **C** $6^2 - 1$
- ☒ **D** $2^6$

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(c) Binary patterns are manipulated by shifts.

(i) Give the result of applying a logical shift right by two to the binary pattern 0101 1100

**(1)**

(ii) Give the result of applying an arithmetic shift right by three to the binary pattern 1100 0101

**(1)**

# 1. Question 2 · part d.i · 2 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part d.i only.

(d) Binary patterns can be interpreted as signed or unsigned integers.

(i) Convert the denary unsigned integer 60 to 8-bit binary.

(2)

(ii) Here is a binary bit pattern for a signed integer in sign and magnitude format.

1001 0110

Convert the binary bit pattern to denary.

Be sure to include a sign symbol in your answer.

(2)

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6

P 7 2 9 3 7 A 0 6 2 4

## 1. Question 2 · part d.i · 2 marks · Working and answer

Working and answer

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# 2. Question 3 · part e.ii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part e.ii only.

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**3** Isra works in an office building where she has access to different types of network.

(a) Isra uses different types of networks for different tasks.

(i) Name the type of network she uses to print a document on the printer in the office next door.

(1)

(ii) Name the type of network she uses to order stationery from an online supplier.

(1)

(b) Identify the measurement used for network data speeds.

(1)

- ☒ **A** Megabits per second
- ☒ **B** Megapixels per second
- ☒ **C** Mebicycles per second
- ☒ **D** Mebihertz per second

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## 2. Question 3 · part e.ii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part e.ii only.

(c) Isra uses her tablet computer and smartphone to access email.

She wants to set up a new email account.

State the email protocol she should use.

Justify your choice.

(3)

Email protocol

Justification

(d) The table describes characteristics of different network topologies.

Complete the table to match each characteristic to **one** network topology.

(4)

| Characteristic | Bus | Ring | Star | Mesh |
|---|---|---|---|---|
| Network performance degrades as more devices are added. | ☒ | ☒ | ☒ | ☒ |
| All devices are directly connected to all others. | ☒ | ☒ | ☒ | ☒ |
| Each node has a physical attachment to a routing device. | ☒ | ☒ | ☒ | ☒ |
| There are no data collisions, because packets travel in the same direction. | ☒ | ☒ | ☒ | ☒ |

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## 2. Question 3 · part e.ii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part e.ii only.

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(e) Information sent across networks is represented in bit patterns.

(i) The bit pattern 1101 0001 uses sign and magnitude representation.

Convert this bit pattern to a denary number.

(2)

(ii) Convert the denary number 75 to 8-bit binary.

(2)

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## 2. Question 3 · part e.ii · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part e.ii only.

(iii) The addition of these two 8-bit binary patterns generates an error condition.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | |
| 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | + |
| 0 | 0 | 1 | 0 | 1 | 1 | 0 | 1 | |

Explain this error condition.

(2)

(Total for Question 3 = 16 marks)

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Working and answer

# 3. Question 4 · part a.vi · 1 mark

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Further practice · Attempt part a.vi only.

**4** Ibrahim works in the music industry.

(a) He is learning how sound is represented in binary.

**Figure 3** illustrates this process.

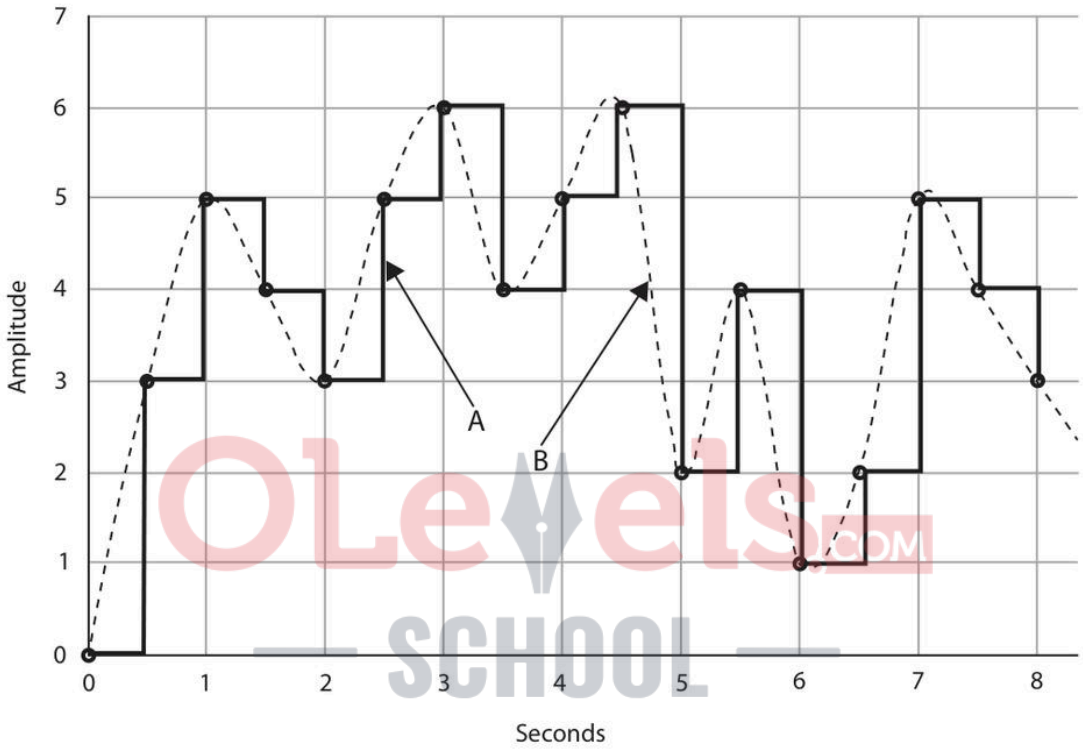



**Figure 3**

(i) State what the solid line labelled **A** represents.

**(1)**

(ii) State what the dashed line labelled **B** represents.

**(1)**

(iii) Give the sampling frequency, including the correct units.

**(1)**

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**10**

P 6 1 8 8 4 R R A 0 1 0 2 0

## 3. Question 4 · part a.vi · 1 mark

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Further practice · Attempt part a.vi only.

(iv) Give **one** impact of increasing the sampling frequency.

**(1)**

(v) The bit-depth measures the resolution of the sound sample.

State the minimum bit-depth needed to store the sound sampled in **Figure 3**.

Justify your answer.

**(2)**

Bit-depth

Justification

(vi) Give the value in binary of the audio sample at the 4th second.

**(1)**

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## 3. Question 4 · part a.vi · 1 mark

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Further practice · Attempt part a.vi only.

(b) Ibrahim wants to store a music file.

- The sampling frequency is 47 kHz.
- The bit-depth is 64
- The analogue signal lasts for 819 seconds.

Construct an expression to show how the file size, in bytes, is calculated.

You do **not** need to do the calculation.

**(4)**

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(c) Ibrahim wants to upload a music file to his online storage. The file has been compressed using a lossy algorithm.

Give **one** drawback of using a lossy algorithm for this purpose.

**(1)**

**(Total for Question 4 = 12 marks)**

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## 3. Question 4 · part a.vi · 1 mark · Working and answer

Working and answer

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## 4. Question 1 · part a.i · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part a.i only.

**1** Computers use binary to represent and store data.

(a) Binary and hexadecimal notation can be used to represent numbers.

(i) Convert the denary number 77 to 8-bit binary.

**(2)**

(ii) Convert the denary number –126 to 8-bit binary using two's complement.

**(2)**

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(iii) Convert the binary pattern 11000110 to hexadecimal.

**(2)**

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**2**

P 6 9 2 9 7 A 0 2 2 0

## 4. Question 1 · part a.i · 2 marks · Working and answer

Working and answer

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## 5. Question 2 · part a.i · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part a.i only.

**2** Computers use binary to represent and store data.

(a) The denary number 78 is the ASCII code for the character **N**.

(i) Convert the denary number 78 to 8-bit binary.

**(2)**

(ii) Identify the number of characters that can be represented using standard ASCII.

**(1)**

- ☒ **A** 64
- ☒ **B** 128
- ☒ **C** 256
- ☒ **D** 512

(iii) Explain **one** reason for using Unicode rather than ASCII to encode languages other than English.

**(2)**

(b) Convert the denary number –43 to 8-bit binary using sign and magnitude representation.

**(2)**

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## 5. Question 2 · part a.i · 2 marks · Working and answer

Working and answer

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## 6. Question 3 · part a.iii · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part a.iii only.

**3** Computers use binary to represent and store data.

(a) Two images are stored.

- Image 1 has a colour depth of 8 bits.
- Image 2 has a colour depth of 16 bits.

(i) State the number of colours that can be represented in Image 1.

**(1)**

(ii) Image 2 is 1920 pixels wide by 1080 pixels high.

Construct an expression to show the file size of the image in mebibytes.

You do **not** need to do the calculation.

**(3)**

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(iii) One pixel in Image 1 is represented by the unsigned binary integer

1100 1010

Identify the denary equivalent.

**(1)**

- ☒ **A** 54
- ☒ **B** 191
- ☒ **C** 202
- ☒ **D** –74

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## 6. Question 3 · part a.iii · 1 mark

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Further practice · Attempt part a.iii only.

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(b) A register holds the value 1100 0001 as an 8-bit binary value in two's complement.

(i) Convert the two's complement binary integer 1100 0001 to denary.

**(2)**

(ii) Give the result of applying an arithmetic shift left by one to the binary pattern 1100 0001

**(1)**

(iii) The value 1111 1111 is added to the value 1100 0001 stored in the register.

An error occurs.

Explain this error.

**(2)**

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## 6. Question 3 · part a.iii · 1 mark · Working and answer

Working and answer

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